Work Integrated Learning (WIL) Report

Incident Response Lab – Phishing-to-Ransomware Attack

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Project Duration: 14 November 2025 – 12 January 2026

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1. Introduction

The objective of this Work Integrated Learning (WIL) project was to design, implement, and analyse a controlled cybersecurity incident response simulation within a virtual lab environment. The project focused on simulating a phishing attack that escalates into a benign ransomware-styled payload, followed by detection, logging, and forensic analysis. The aim was to gain practical exposure to real-world cybersecurity concepts, including social engineering, intrusion detection, system logging, incident analysis, and reporting.

This project was conducted entirely in a simulated and isolated environment to ensure safety, legality, and ethical compliance. No real users or production systems were affected. The project emphasizes learning outcomes rather than exploitation, with all malicious actions being simulated and reversible.

2. Background and Context

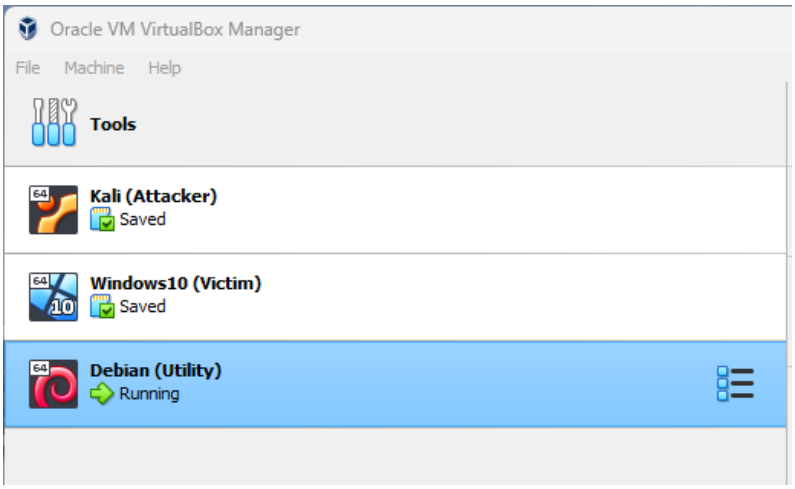
Phishing attacks remain one of the most common initial access vectors used by attackers to compromise systems. Once a user interacts with a malicious email or link, attackers often deploy payloads such as ransomware, leading to data encryption, operational disruption, and financial loss. This project was initiated to understand how such attacks occur, how they can be detected, and how incident response teams analyse and document security incidents. A small-scale lab environment was chosen to reflect a typical organizational setup with a user workstation, an attacker machine, and a utility server for email and monitoring services. Due to limited hardware resources (8 GB RAM and 256 GB storage), the lab design intentionally avoided complex enterprise components such as domain controllers and file servers. Instead, the focus was placed on core attack and detection mechanisms relevant to learning outcomes.

3. Project Environment and Architecture

The virtual lab consisted of three virtual machines hosted on Virtual Box:

- **Windows 10 Pro (Victim Machine)**
  Simulated an end-user simulation where phishing emails were and the ransomware simulation was executed.
- **Kali Linux (Attacker)**
  Hosted GoPhish for phishing campaign creation and delivery.

- **Debian 12 (Utility)**
  Hosted Postfix and Dovecot for internal email services, Suricata IDS for networking and tcpdump for packet capture.



Networking was configured primarily using a Host-based network, with NAT used where required. This design allows isolated communication between virtual machines while maintaining safety.

Host-only Network:

Kali IP Address Configurations:

```
┌──(root㉿attacker)-[~]
└─# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.110  netmask 255.255.255.0  broadcast 0.0.0.0
        ether 08:00:27:53:24:4c  txqueuelen 1000  (Ethernet)
        RX packets 1  bytes 60 (60.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.3.15  netmask 255.255.255.0  broadcast 10.0.3.255
        inet6 fe80::a00:27ff:feb6:8401  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:b6:84:01  txqueuelen 1000  (Ethernet)
        RX packets 649  bytes 209348 (204.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 642  bytes 122057 (119.1 KiB)
        TX errors 0  dropped 1 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 466  bytes 837713 (818.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
```

Debian IP Address Configurations:

```
uendjipa@Utility: ~
uendjipa@Utility:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:c5:22:11 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fec5:2211/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:90:60:c9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixroute enp0s8
       valid_lft 86364sec preferred_lft 86364sec
    inet6 fe80::a00:27ff:fe90:60c9/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Windows IP Address Configurations:

```
Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::9f8:2061:af6f:1341%5
   IPv4 Address. . . . . . . . . . . : 192.168.56.105
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::c4fd:bdc:ff59:5608%10
   IPv4 Address. . . . . . . . . . . : 10.0.3.15
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.0.3.2
```

4. Scope of the Project

Included in Scope:

- Design and deployment of a phishing campaign using GoPhish
- Creation of phishing email templates and landing pages
- Development of a benign ransomware simulation script
- Execution of phishing and payload simulation on a test account
- Configuration of logging and monitoring tools

- Capture and analysis of logs and network traffic
- Incident timeline reconstruction and reporting

Excluded from scope:

- Real ransomware or destructive malware
- Attacks against real users or live systems
- Enterprise-scale infrastructure (Domain Controllers, SIEM platforms)
- Automated lateral movement or privilege escalation

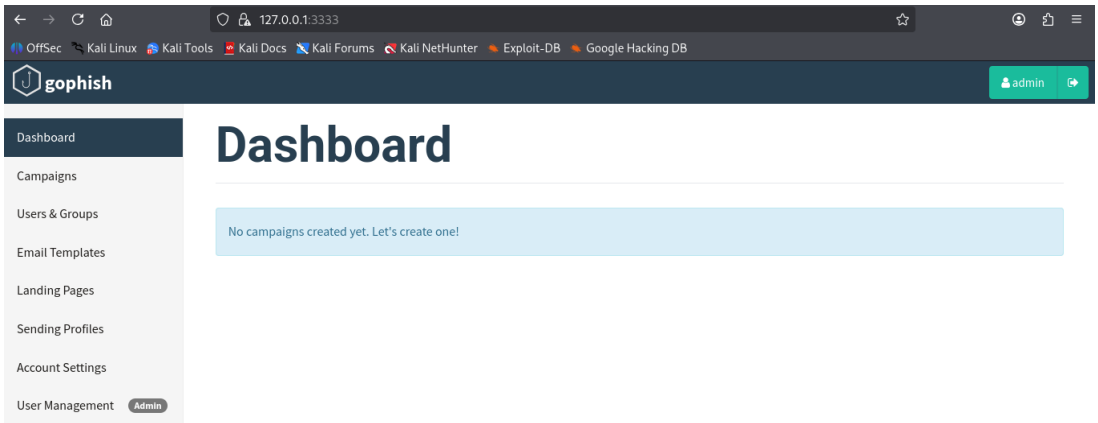
5. Roles and Responsibilities

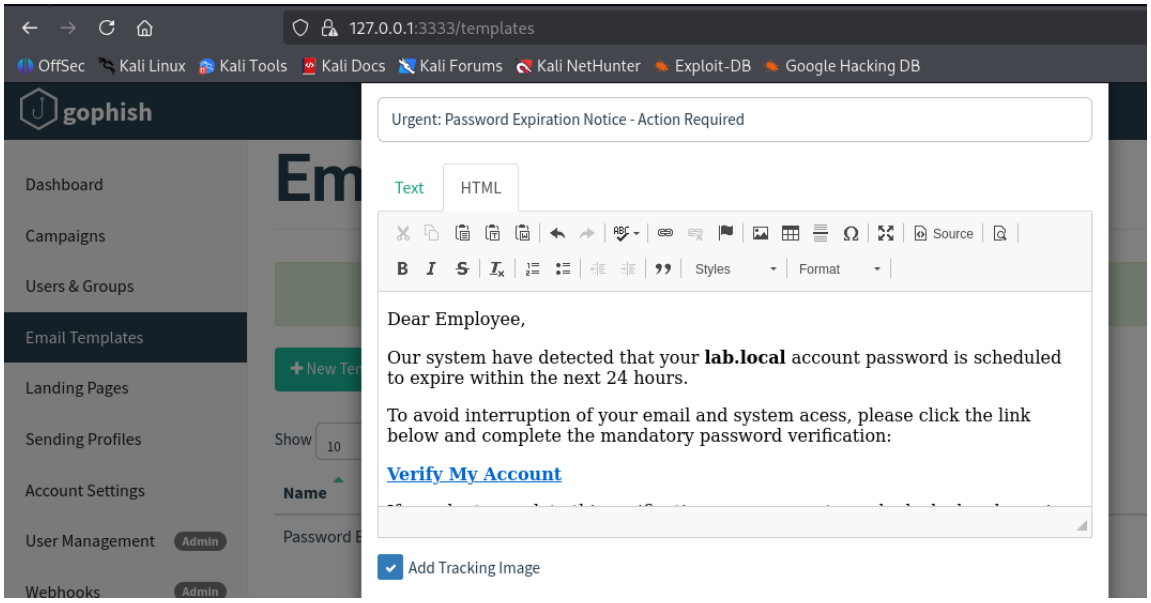
I was solely responsible for the design, implementation, execution, monitoring, and documentation of the project. This included configuring all virtual machines, developing scripts, running simulations, troubleshooting issues, collecting evidence, and compiling reports. The project was completed independently as a learning exercise aligned with WIL requirements.

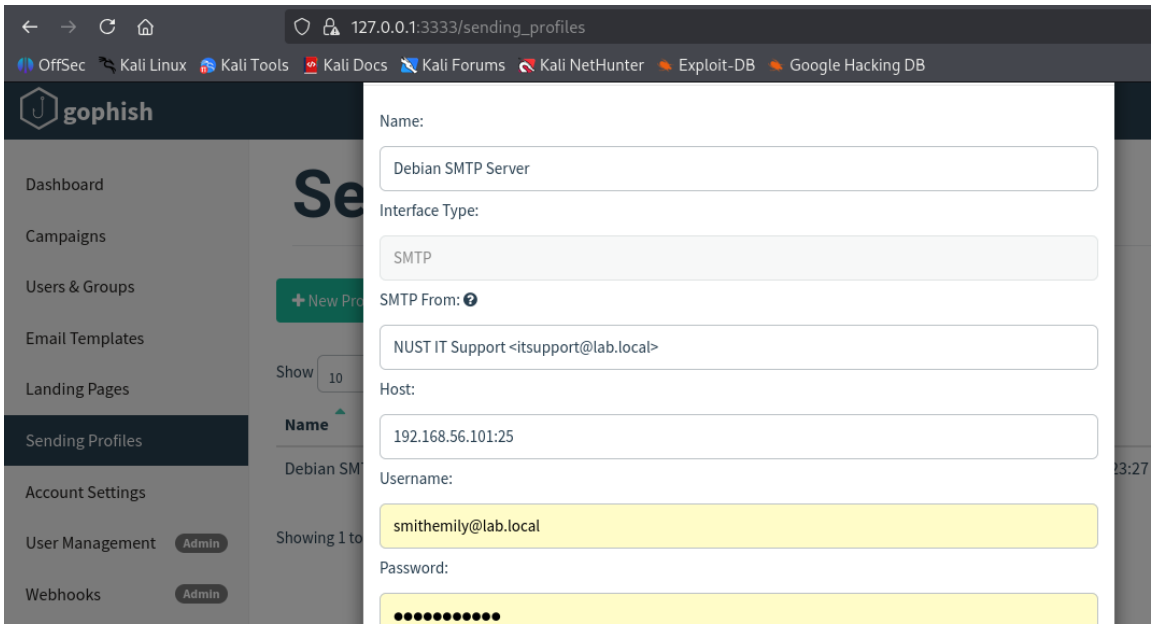
6. Methodology/Tools

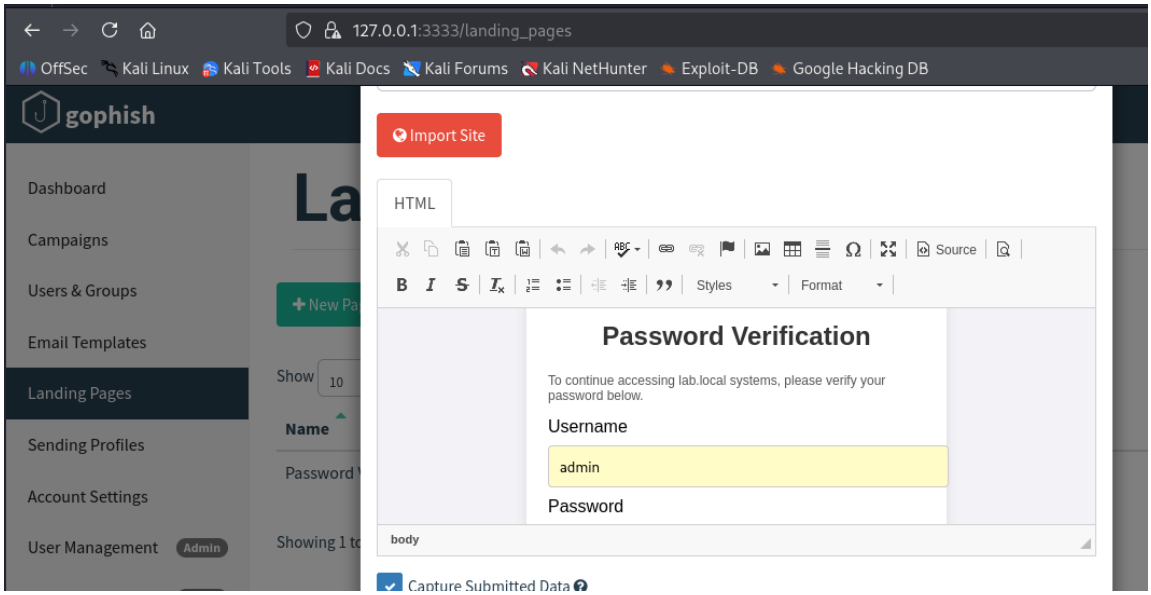
6.1 Attack Simulation

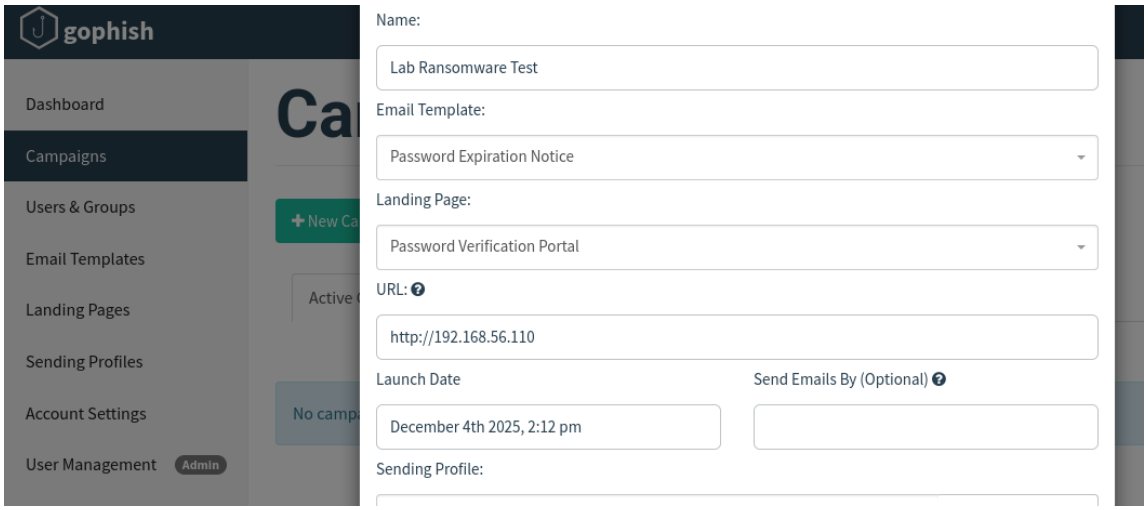
- GoPhish was used to create phishing email templates, landing pages, and sending profiles.
- A test phishing campaign was executed against a single internal test account.
- User interaction was simulated by clicking the phishing link.

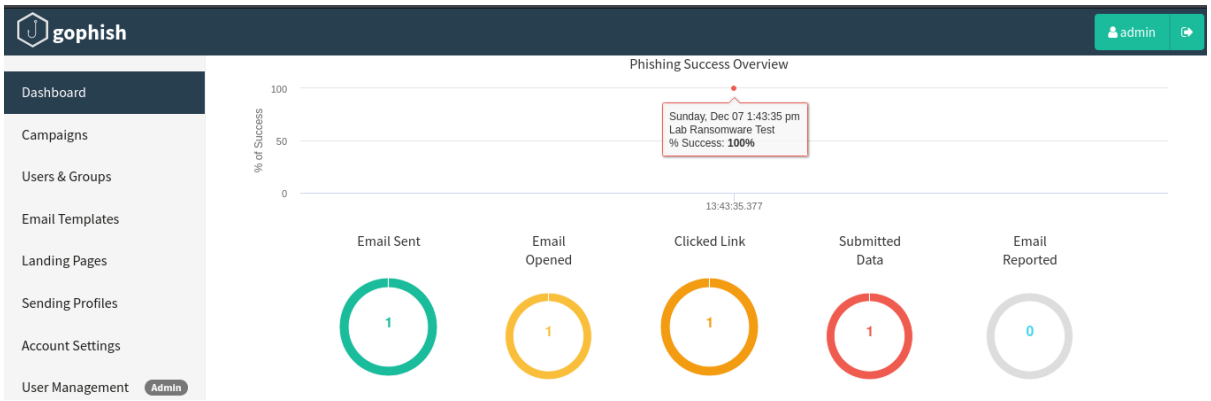



127.0.0.1:3333/templates
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
gophish
Dashboard
Campaigns
Users & Groups
Email Templates
Landing Pages
Sending Profiles
Account Settings
User Management
Admin
Webhooks
Admin
Urgent: Password Expiration Notice - Action Required
Text
HTML
Source
Styles
Format
Dear Employee,
Our system have detected that your lab.local account password is scheduled to expire within the next 24 hours.
To avoid interruption of your email and system acess, please click the link below and complete the mandatory password verification:
Verify My Account
Add Tracking Image


127.0.0.1:3333/sending_profiles
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
gophish
Dashboard
Campaigns
Users & Groups
Email Templates
Landing Pages
Sending Profiles
Account Settings
User Management
Admin
Webhooks
Admin
Name:
Debian SMTP Server
Interface Type:
SMTP
SMTP From:
NUST IT Support <itsupport@lab.local>
Host:
192.168.56.101:25
Username:
smithemily@lab.local
Password:
●●●●●●●●●●●







6.2 Payload Simulation

- A PowerShell-based benign ransomware simulation script was developed.
- The script encrypted files in selected user directories (Documents and Desktop) and generated a ransom note.
- The script was executed manually to simulate post-phishing payload deployment.

Powershell Benign Script:

*simulate_ransom - Notepad

File Edit Format View Help

```
Write-Host "Starting benign ransomware simulation..."

# SET TARGET DIRECTORIES

$DocDir = "$env:USERPROFILE\Documents"
$DeskDir = "$env:USERPROFILE\Desktop"


# CREATE RANSOMWARE FOLDERS

$DocRansomDir = "$DocDir\RANSOM_LOCKED"
$DeskRansomDir = "$DeskDir\RANSOM_LOCKED"

foreach ($dir in @($DocRansomDir, $DeskRansomDir)) {
    if (!(Test-Path $dir)) {
        New-Item -ItemType Directory -Path $dir | Out-Null
     }
}
```

Ln 29, Col 37 | 100% | Windows (CRLF) | UTF-8

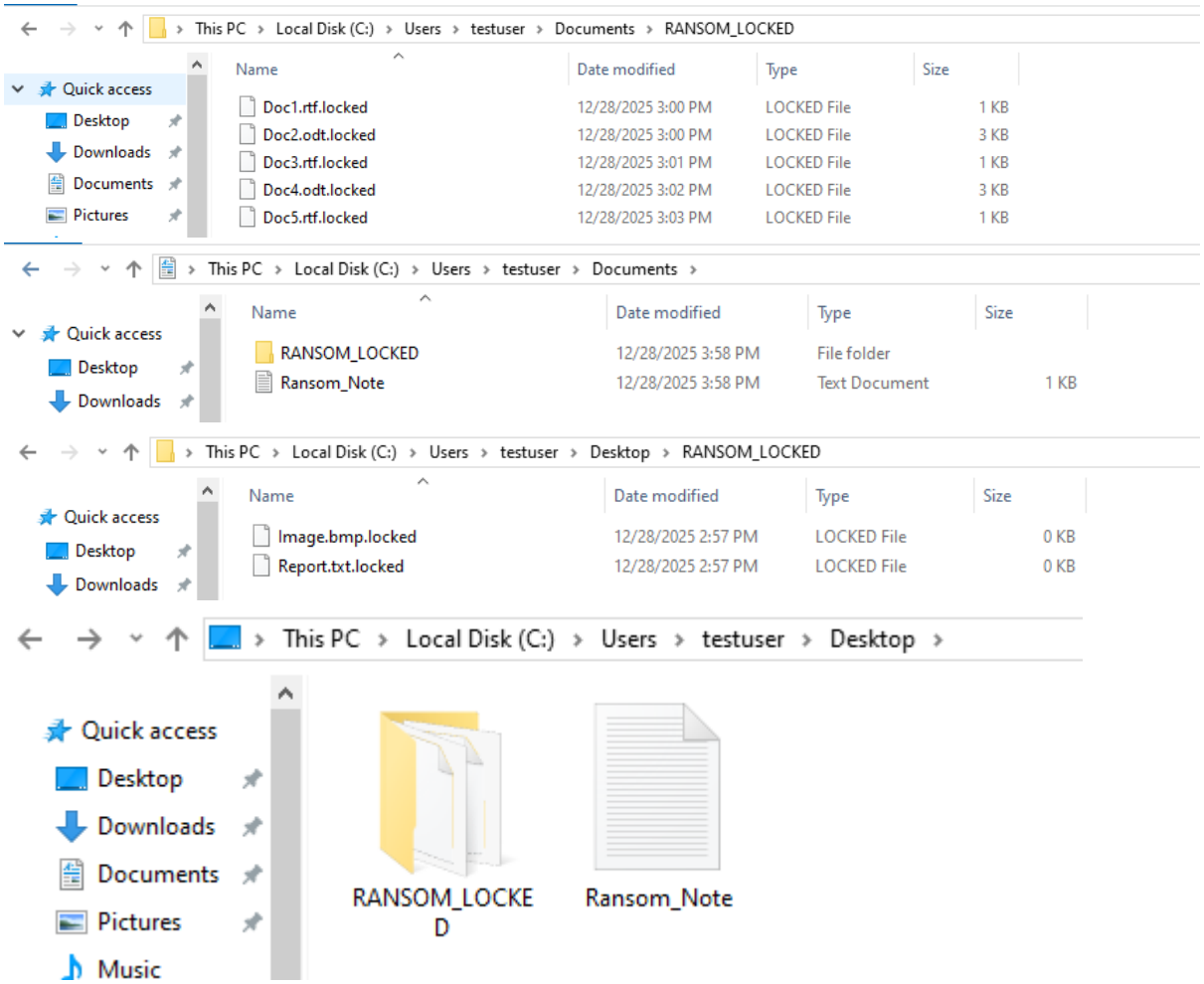
```
# Function to lock files

function Lock-Files {
   param (
      [string]$SourceDir,
      [string]$OutputDir
)


Get-ChildItem -Path $SourceDir -File | ForEach-Object {
    $newName = "$($_.Name).locked"
    Rename-Item -Path $_.FullName -NewName $newName -Force
    Move-Item -Path "$SourceDir\$newName" -Destination $OutputDir -Force
    Write-Host "[+] Locked: $($_.Name) from $SourceDir"
   }
}

# Lock Documents + Desktop files
Lock-Files -SourceDir $DocDir -OutputDir $DocRansomDir
```

```
Lock-Files -SourceDir $DocDir -OutputDir $DocRansomDir
Lock-Files -SourceDir $DeskDir -OutputDir $DeskRansomDir

# DROP RANSOM NOTE INTO BOTH FOLDERS
$RansomNote = @"
Your important files have been encrypted.
This is a benign simulation for cybersecurity training.
No real data has been harmed.

- Security Team
"@

$DocNote = "$DocDir\Ransom_Note.txt"
$DeskNote = "$DeskDir\Ransom_Note.txt"

$RansomNote | Out-File -FilePath $DocNote -Encoding UTF8 -Force
$RansomNote | Out-File -FilePath $DeskNote -Encoding UTF8 -Force

Write-Host "Simulation complete. Ransom notes created in Documents and Deskto
```

Powershell Output:

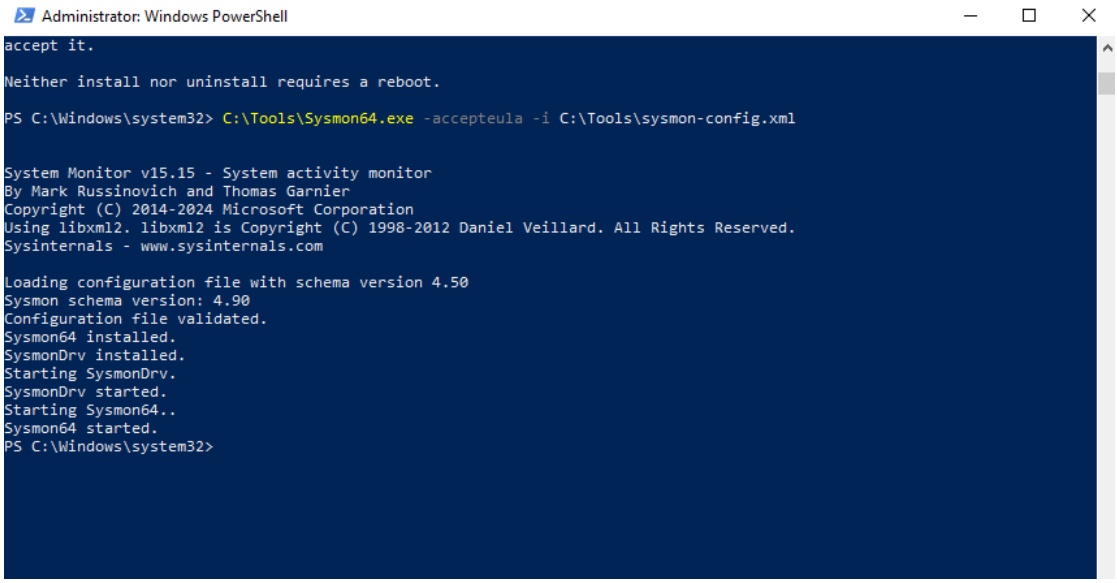
```
PS C:\Users\testuser> powershell -ExecutionPolicy Bypass -File C:\Tools\simulate_ransom.ps1
Starting benign ransomware simulation...
[+] Locked: doc1.txt from C:\Users\testuser\Documents
[+] Locked: doc2.txt from C:\Users\testuser\Documents
[+] Locked: doc3.txt from C:\Users\testuser\Documents
[+] Locked: doc4.txt from C:\Users\testuser\Documents
[+] Locked: doc5.txt from C:\Users\testuser\Documents
[+] Locked: Image.bmp from C:\Users\testuser\Desktop
[+] Locked: Report.txt from C:\Users\testuser\Desktop
Simulation complete. Ransom notes created in Documents and Desktop.
PS C:\Users\testuser>
```

Encrypted Files and Ransom Note Creation:

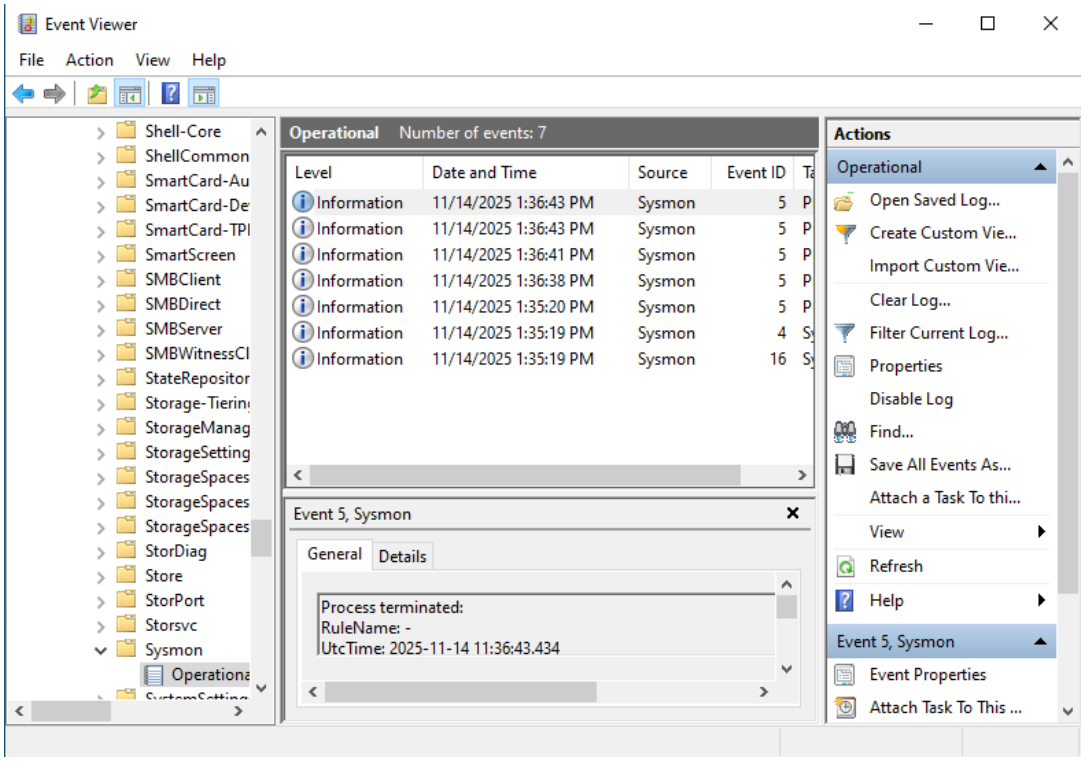


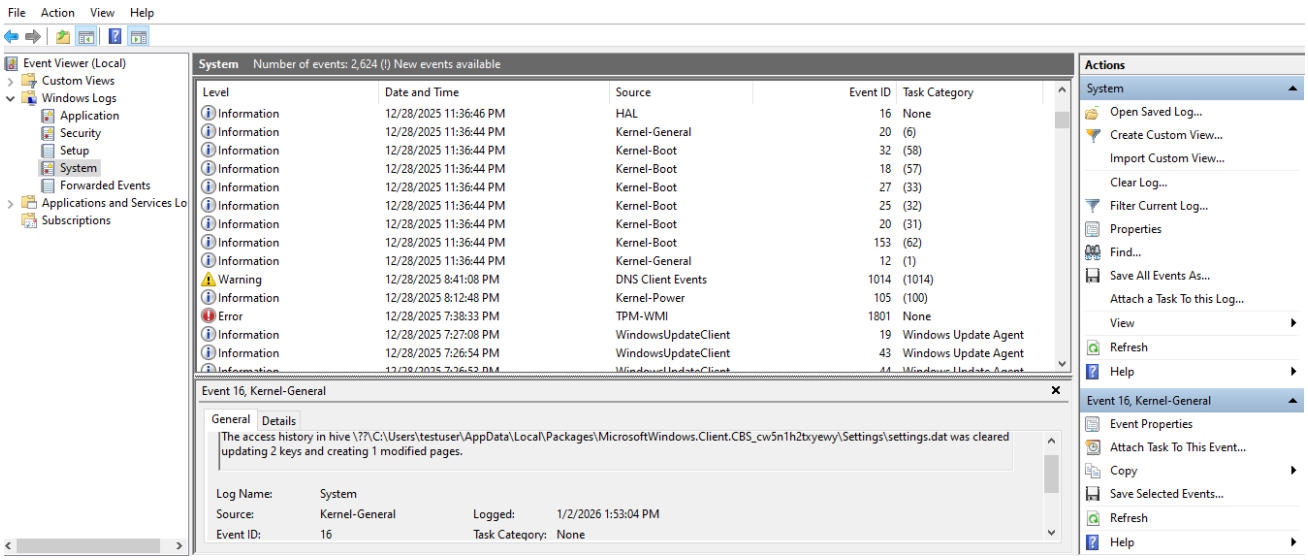
6.3 Logging and Monitoring

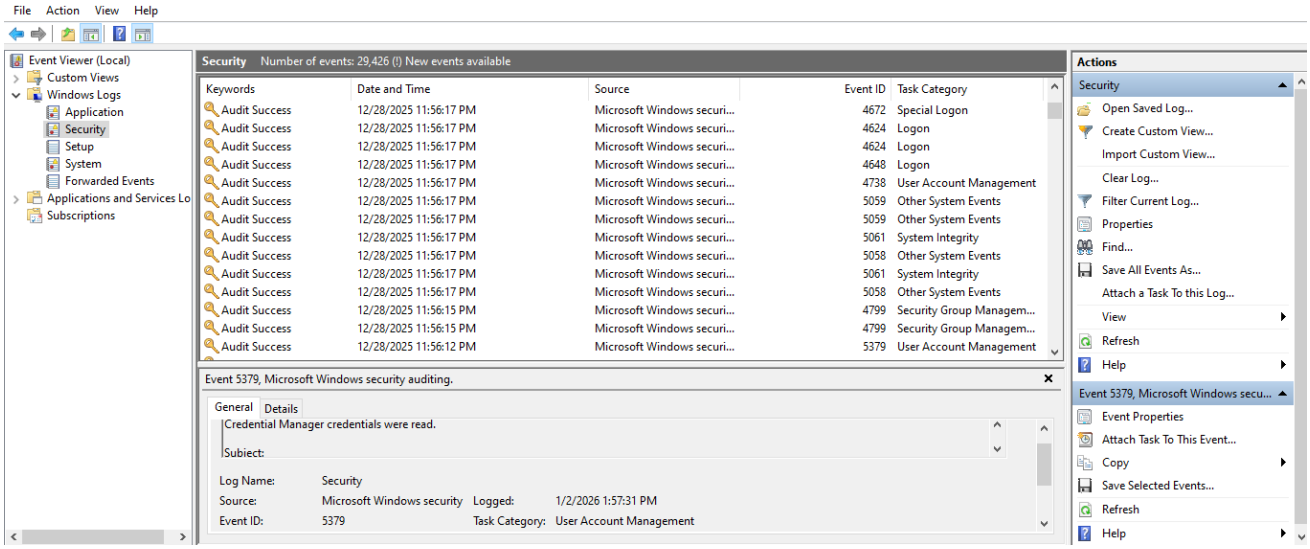
- Sysmon was installed on the Windows machine for host-based logging.
- Suricata IDS was deployed on Debian for network monitoring.
- Postfix and Dovecot logs were used to track email delivery.
- tcpdump was used to capture PCAP files during attack execution.

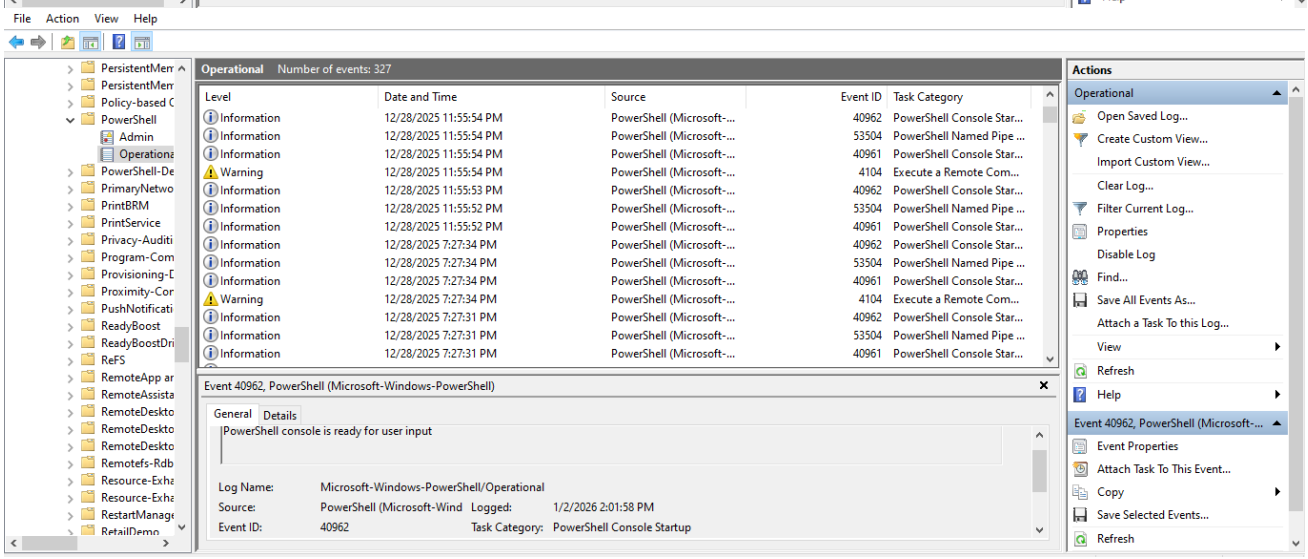


Windows Event Viewer:







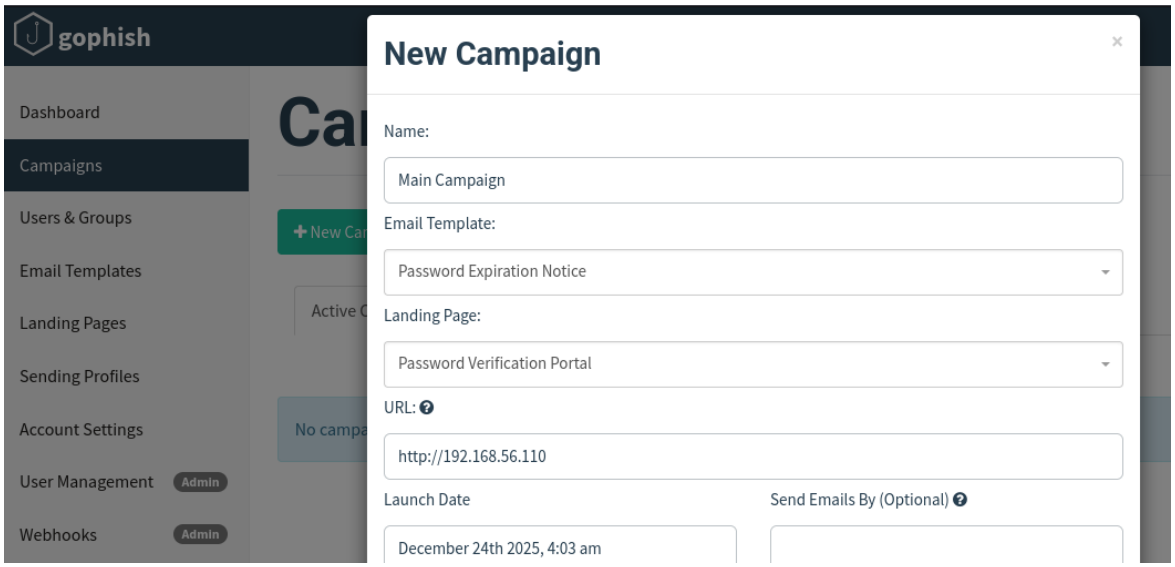


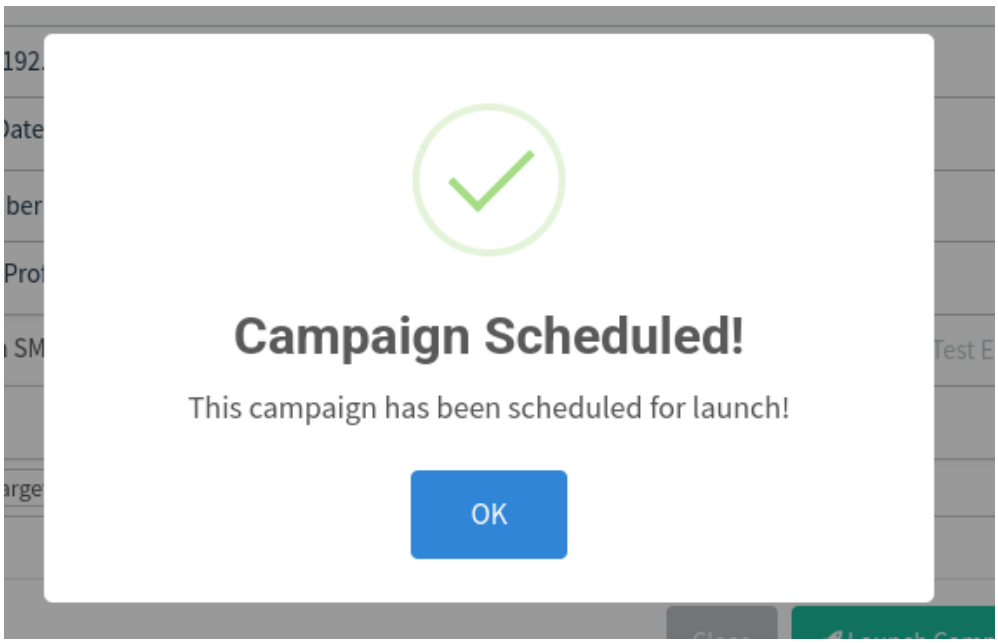
7. Implementation and Execution

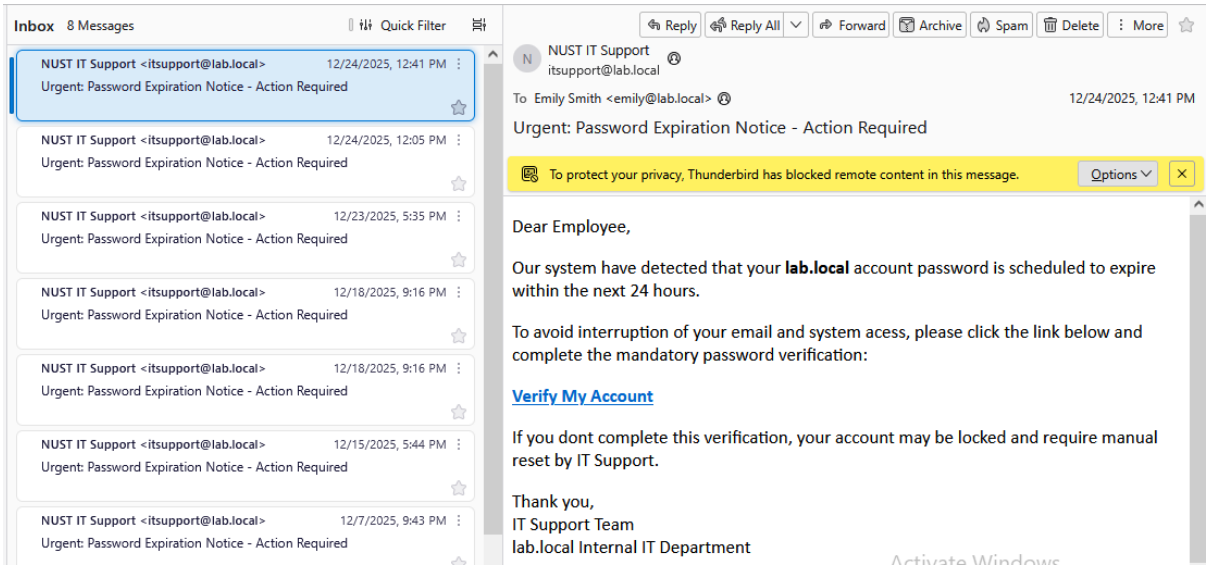
The project progressed through multiple phases:

1. Lab setup and system hardening
2. Phishing content creation and testing
3. Email delivery validation
4. Payload execution and file locking
5. IDS monitoring and packet capture
6. Log collection and analysis

A pilot campaign was first executed to validate configurations, followed by a controlled main campaign.





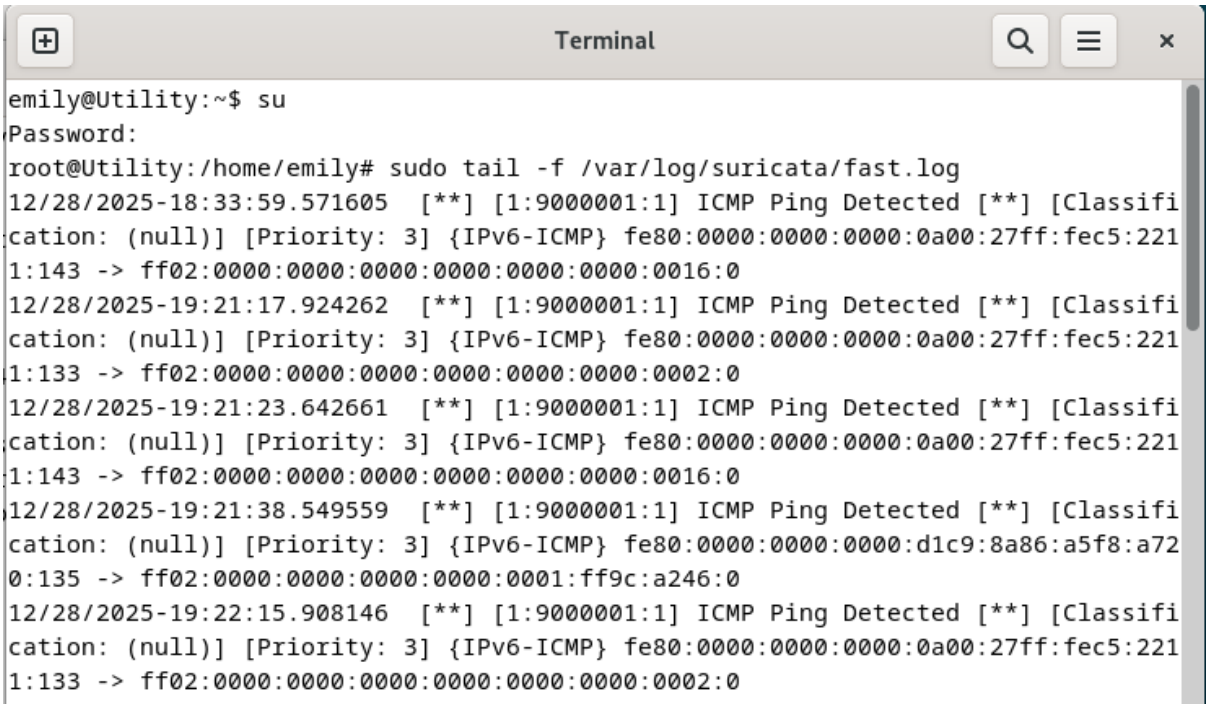


8. Analysis and Findings

Captured logs confirmed:

- Successful phishing email delivery
- User interaction with the phishing link
- Execution of the ransomware simulation script
- File encryption within the user directories

Suricata successfully logged ICMP traffic but did not consistently log SMTP and HTTP traffic.



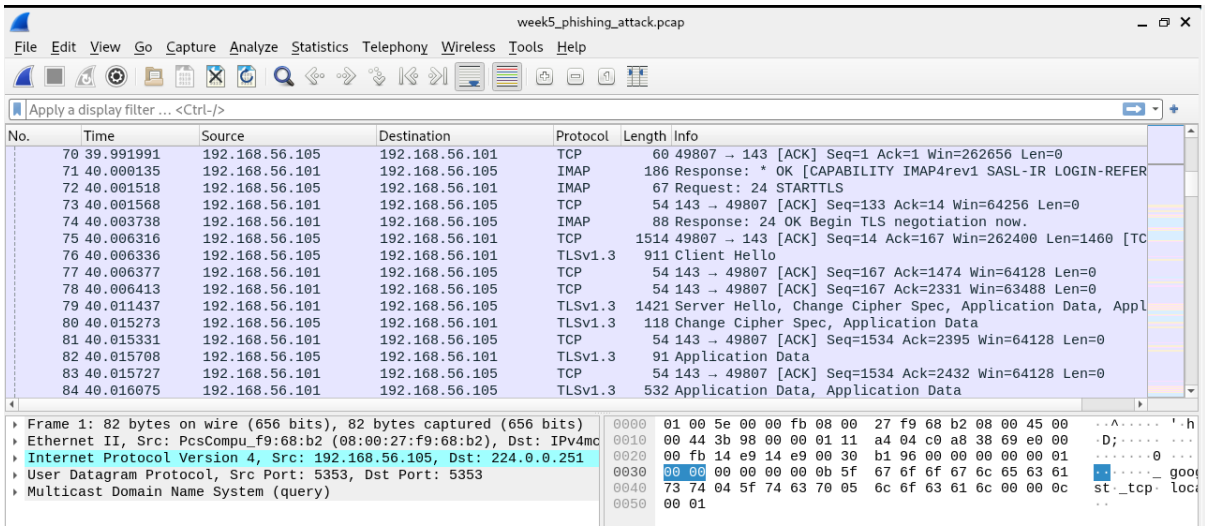
This limitation was attributed to:

- Interface monitoring constraints
- Hardware limitations preventing all VMs from running simultaneously

Despite these limitations, tcpdump PCAPs and application logs provided sufficient evidence to reconstruct the incident timeline.

```
Terminal
emily@Utility:~$ su
Password:
root@Utility:/home/emily# sudo journalctl -u postfix > postfix.log
root@Utility:/home/emily# sudo journalctl -u dovecot > dovecot.log
root@Utility:/home/emily#
```

```
Terminal
emily@Utility:~$ su
Password:
root@Utility:/home/emily# sudo journalctl -u dovecot -f
Dec 28 23:42:54 Utility systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3
 email server...
Dec 28 23:42:55 Utility dovecot[647]: master: Dovecot v2.3.19.1 (9b53102964) sta
rting up for imap, pop3 (core dumps disabled)
Dec 28 23:42:55 Utility systemd[1]: Started dovecot.service - Dovecot IMAP/POP3
email server.
Dec 28 23:55:04 Utility dovecot[697]: imap-login: Login: user=<emily>, method=PL
AIN, rip=192.168.56.105, lip=192.168.56.101, mpid=2714, TLS, session=<KYTXLQpHkM
LAqDhp>
Dec 28 23:55:05 Utility dovecot[697]: imap-login: Login: user=<emily>, method=PL
AIN, rip=192.168.56.105, lip=192.168.56.101, mpid=2716, TLS, session=<KnDlLQpHj8
LAqDhp>
Dec 29 00:25:14 Utility dovecot[697]: imap(emily)<2716><KnDlLQpHj8LAqDhp>: Disco
nnected: Inactivity - no input for 1800 secs in=173 out=835 deleted=0 expunged=0
 trashed=0 hdr_count=0 hdr_bytes=0 body_count=0 body_bytes=0
```



9. Challenges and Limitations

- Limited system resources restricted simultaneous VM operation
- IDS visibility gaps due to network interface selection
- Some logs could not capture events in real time

These challenges reflect realistic constraints often faced in small-scale environments and were documented as part of the learning process.

10. Learning Outcomes

Through this project, I gained practical experience in:

- Phishing attack design and execution
- Incident detection and logging
- Network traffic analysis
- Host-based and network-based monitoring
- Incident documentation and reporting
- Ethical and controlled malware simulation

11. Conclusion

This WIL project successfully demonstrated a full incident response lifecycle using a controlled phishing-to-ransomware simulation. Despite hardware and environmental constraints, the project achieved its objectives by capturing meaningful evidence, analysing attack behaviour, and producing comprehensive documentation. The experience reinforced the importance of proper monitoring, logging, and system design in cybersecurity operations and provided valuable hands-on exposure to real-world incident response practices.

12. Recommendations

Future enhancements could include:

- Centralized log management (SIEM)
- Improved IDS interface coverage
- Additional simulated users
- Automation of response actions

13. Appendices

- Weekly progress logs
- Scripts and configurations
- Screenshots and PCAP files
- Sample logs and outputs